

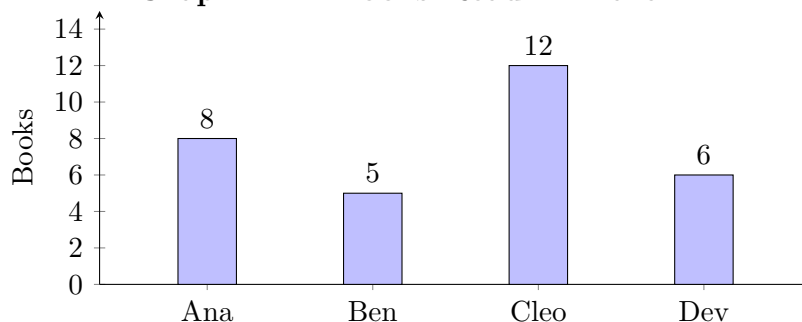
Graphs, Line Plots, and Shape Sorting

Grade 2–3 Math

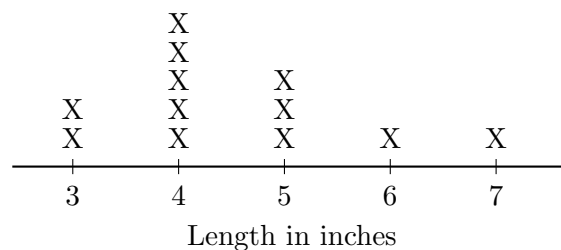
Reading a bar graph, reading a line plot, comparing and measuring data, and sorting shapes by their sides and corners. Use the three displays below.

- **Look first, then decide.** Some questions ask you to read one bar or one stack of X marks. Others ask you to add, to compare, or to sort shapes. Decide which one the question wants before you write.
- **Show how you got it.** Write the numbers you used, not just the answer.
- **Sides and corners.** A shape with 4 sides always has 4 corners. A square needs 4 equal sides *and* 4 square corners.

Graph A — Books Read in March



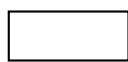
Line Plot B — Pencil Lengths in Ms. Ruiz's Cup (inches)



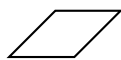
Shape Set C (no measurements shown — use the sides and corners you can see)



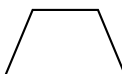
P



Q



R



S



T



U

1. **Graph A.** How many books did Cleo read in March?

Answer: _____

2. Graph A. How many books did all four students read altogether?

Answer: _____

3. Line Plot B. How many pencils are exactly 4 inches long?

Answer: _____

4. Line Plot B. How many pencils are in the cup altogether?

Answer: _____

5. Shape Set C. Sort the six shapes by how many sides each one has, then fill in the chart.

Number of sides	How many shapes
3 sides	
4 sides	
5 sides	
6 sides	

6. Line Plot B. How much longer is the longest pencil than the shortest pencil?

Answer: _____ inches

7. Graph A. How many more books did Cleo read than Ben?

Answer: _____

8. Shape Set C. Dev says, “Every shape with 4 sides is a square.” Use shapes P, Q, R and S to explain why Dev is wrong. Name one four-sided shape from the set that is *not* a square, and tell what a square needs that it does not have.

9. Line Plot B. Which pencil length is the most common one in the cup?

Answer: _____ inches

10. Line Plot B. How many pencils are longer than 4 inches? Add up every X mark to the right of the 4.

Answer: _____

11. Mia draws a square with each side 6 cm long. She also draws a rectangle that is 8 cm long and 3 cm wide.

How much greater is the distance around Mia's square than the distance around her rectangle?

Answer: _____ cm

12. Synthesis challenge — Shape Set C. Ms. Ruiz draws one more shape and adds it to Shape Set C. The new shape has 4 sides, and all 4 of its sides are the same length.

(a) There are now 7 shapes in the set. How many of them have exactly 4 sides?

(a) _____

(b) Ben says, "Now more than half of the shapes have 4 sides." Is Ben right? Use numbers to explain.